MORGAN BAUER

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Education

University of Pennsylvania — Philadelphia, PA
Candidate for Master of Science in Engineering in Robotics — GRASP Laboratory — *Expected May 2027*

- **Relevant Coursework:**
  - Big Data Analytics
  - Engineering Entrepreneurship
  - Machine Learning
  - Computer Vision and Computational Photography
  - Introduction to Robotics
  - Mathematics for Robotics
  - Engineering Economics
  - Learning for Robotics

Eckerd College — St. Petersburg, FL
Candidate for Bachelor of Science in Computer Science and Mathematics — *Conferred May 2025*

- **Relevant Coursework:**
  - Abstract Algebra I
  - Bioinformatics
  - Calculus I & II
  - Cognitive Linguistics (audit)
  - Computer Architecture
  - Computer Programming Concepts
  - Cybersecurity
  - Data Science Fundamentals
  - Data Structures
  - Differential Equations
  - Digital Image Processing
  - Electronics
  - Evolutionary Computation
  - GIS for Environmental Studies
  - GUI Design (audit)
  - Intermediate Programming
  - Intro to Mathematical Thinking
  - Linear Algebra
  - Machine Learning
  - Number Theory
  - Partial Differential Equations
  - Probability & Statistics I
  - Programming Languages
  - Quantum Physics I
  - Real Analysis I
  - Software Capstone
  - Theory of Computing
  - Translators & Compilers
  - Waves and Relativity

Research Experience

Eckerd College — St. Petersburg, FL
Deep Learning for Musical Accompaniment Generation — Undergraduate Thesis — *July 2024 – May 2025*

- Designed and implemented a Transformer architecture using PyTorch to generate musical accompaniment from a solo input stream, proving its viability for real-time applications (Advised by Dr. Michael Hilton).

Computer Science Research Assistant — *June 2023 – May 2025*

- Developing a new version of DARWIN (Digital Analysis and Recognition of Whale Images on a Network) which serves hundreds of marine scientists globally. Contributed to an updated user interface for fin tracing and contour extraction.

Ford Apprentice Scholar — Module-Lattice-Based Key-Encapsulation Mechanism (FIPS 203) — *April 2023 – May 2025*

- Nominated, selected, and funded to develop research and pedagogy skills. Analyzed and implemented post-quantum encryption standards in Python.

Undergraduate Mathematics Researcher — *December 2022 – May 2025*

- Conducted research on the combinatorial properties of ordered trees, specifically investigating the Steiner k-Wiener index.

Publications & Presentations

- **Publication:** Bauer, M. and Copenhaver, K., 2025. A new lower bound for deterministic pop-stack-sorting. *European Journal of Combinatorics*, 124, p. 104046.
- **Invited Talk:** Pop-stack-sorting in a special session on enumerative combinatorics at the 55th Southeastern International Conference on Combinatorics, Graph Theory, and Computing.
- **Presentation:** Analysis and Implementation of Module-Lattice-Based Key-Encapsulation Mechanism Standard (FIPS 203) at the 2024 Eckerd College NAS Student Research Symposium.
- **Presentation:** On the Steiner k-Wiener Index of Ordered Trees at the 2024 Eckerd College NAS Student Research Symposium.

Teaching & Institutional Experience

Eckerd College — St. Petersburg, FL
Tutor in Mathematics — *September 2024 – May 2025*

- **Courses Tutored:** Precalculus, Calculus I, Calculus II.

Computer Science Tutor and Lab Assistant — *January 2023 – May 2025*

- **Courses Tutored:** Computer Programming Concepts, Intermediate Programming, Data Structures, Discrete Structures, Python for the Biological Sciences.

Teaching Assistant — How To Design an Escape Room — *May 2024 – September 2024*

- Lectured on cryptographic concepts and introduced micro:bits hardware/software.

Teaching Assistant — Introduction to Aniamtronics — *February 2024 – May 2024*

- Assisted with teaching topics including programming, electronics, and soldering.

Student Representative — Computer Policy Group — *August 2023 – May 2025*

- Served as the sole student member making institutional decisions regarding campus technology policy and infrastructure.

Other Professional Experience

Arizona Cardinals — Glendale, AZ
Training Camp Assistant — Football Analytics — *July 2026 – August 2026*

- Applied machine learning techniques, including graph neural networks, variational autoencoders, and proximal policy optimization, with on-field tracking data; quantify quarterback pass IQ, predict rushing yards, and improve draft preparation.

United States Geological Survey — St. Petersburg, FL
Information Technology Intern — *September 2024 – May 2025*

- Assisted geoscience researchers with Python and R troubleshooting, scientific software configuration, and maintenance on production servers.

Honors & Awards

- **Climate Fellow**, Climate Leaders @ Penn — October 2025
- **Edmund L. Gallizzi Award for Excellence in Computer Science** — May 2025
- **Meacham Mathematics Memorial (M^3) Award** — May 2025
- **Pi Mu Epsilon Florida Gamma Chapter** — May 2024
- **Ford Apprentice Scholar** — April 2023
- **Dean's List** — 2023 – 2025
- **Natural Sciences Achievement Scholarship** — 2022 – 2023
- **Music Performance Scholarship** — 2021 – 2024

Skills

Robotics & Systems:
· ROS, Gazebo, ARMv8 Assembly
· Git, Linux, LaTeX, Feedback Control
· Kinematics, Motion Planning

Data Science & ML:
· PyTorch, TensorFlow, Keras
· NumPy, Jupyter Notebooks

Languages & App Dev:
· Python, R, SQL, Swift
· C#, Java, XCode, Microsoft Office

Interests & Certifications

SCUBA Diving:
· TDI Technical Diver (ANDP)
· SSI Dive Guide, Rescue Diver

Other Outdoors:
· Biking, Hiking, Camping
· Snowboarding, Birding

Creative & Games:
· Woodwinds (especially Tenor Sax)
· Board Games, Reading
· Wildlife Photography